EASY

PROBLEM-SOLVING AND DATA ANALYSIS

Probability and conditional probability

01

10 questions · ~15 min

SAT QUESTION BANK

2026-05-29

Q1 PROBLEM-SOLVING AND DATA ANALYSIS

$$-13, 4, 23$$

A data set of three numbers is shown. If a number from this data set is selected at random, what is the probability of selecting a negative number?

A. 0

B. $\frac{1}{3}$

C. $\frac{2}{3}$

D. 1

Q2 PROBLEM-SOLVING AND DATA ANALYSIS

A store received a shipment of $1{,}000$ MP3 players, 4 of which were defective. If an MP3 player is randomly selected from this shipment, what is the probability that it is defective?

A. 0.004

B. 0.04

C. 0.4

D. 4

Colors of Marbles in a Bag

Color	Number
Red	8
Blue	10
Green	22
Total	40

The table shows the number of different colors of marbles in a bag. If a marble is chosen at random from the bag, what is the probability that the marble will be blue?

A. $\frac{30}{40}$

B. $\frac{22}{40}$

C. $\frac{18}{40}$

D. $\frac{10}{40}$

Voice type	Number of singers
Countertenor	4
Tenor	6
Baritone	10
Bass	5

A total of 25 men registered for singing lessons. The frequency table shows how many of these singers have certain voice types. If one of these singers is selected at random, what is the probability he is a baritone?

A. 0.10

B. 0.40

C. 0.60

D. 0.67

Q5

PROBLEM-SOLVING AND DATA ANALYSIS

-11, -9, 26

A data set of three numbers is shown. If a number from this data set is selected at random, what is the probability of selecting a positive number?

A. 0

B. $\frac{1}{3}$

C. $\frac{2}{3}$

D. 1

Q6

PROBLEM-SOLVING AND DATA ANALYSIS

A bag contains a total of 60 marbles. A marble is to be chosen at random from the bag. If the probability that a blue marble will be chosen is 0.35, how many marbles in the bag are blue?

A. 21

B. 25

C. 35

D. 39

	Live east of the river	**Live west of the river**	**Total**
Less than 40 **years old**	17	11	28
At least 40 **years old**	18	89	107
Total	35	100	135

The table summarizes members of a local organization by age and whether they live east or west of the river. If a member of the organization is selected at random, what is the probability that the selected member is at least 40 years old?

A. $\frac{28}{135}$

B. $\frac{35}{135}$

C. $\frac{100}{135}$

D. $\frac{107}{135}$

Q8 PROBLEM-SOLVING AND DATA ANALYSIS

There are n nonfiction books and 12 fiction books on a bookshelf. If one of these books is selected at random, what is the probability of selecting a nonfiction book, in terms of n ?

A. $\frac{n}{12}$

B. $\frac{n}{n+12}$

C. $\frac{12}{n}$

D. $\frac{12}{n+12}$

Number of High School Students Who
Completed Summer Internships

High school	Year				
	2008	**2009**	**2010**	**2011**	**2012**
Foothill	87	80	75	76	70
Valley	44	54	65	76	82
Total	131	134	140	152	152

The table above shows the number of students from two different high schools who completed summer internships in each of five years. No student attended both schools. Of the students who completed a summer internship in 2010, which of the following represents the fraction of students who were from Valley High School?

A. $\frac{10}{140}$

B. $\frac{65}{140}$

C. $\frac{75}{140}$

D. $\frac{65}{75}$

The table gives the distribution of votes for a new school mascot and grade level for 80 students.

Mascot	Grade level			
	Sixth	**Seventh**	**Eighth**	**Total**
Badger	4	9	9	22
Lion	9	2	9	20
Longhorn	4	6	4	14
Tiger	6	9	9	24
Total	23	26	31	80

If one of these students is selected at random, what is the probability of selecting a student whose vote for new mascot was for a lion?

A. $\frac{1}{9}$

B. $\frac{1}{5}$

C. $\frac{1}{4}$

D. $\frac{2}{3}$